

QKD: Quantization-aware Knowledge Distillation

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Abstract

Quantization and Knowledge distillation (KD) methods are widely used to reduce memory and power consumption of deep neural networks (DNNs), especially for resource-constrained edge devices. Although their combination is quite promising to meet these requirements, it may not work as desired. It is mainly because the regularization effect of KD further diminishes the already reduced representation power of a quantized model. To address this shortcoming, we propose Quantization-aware Knowledge Distillation (QKD) wherein quantization and KD are carefully coordinated in three phases. First, Self-studying (SS) phase fine-tunes a quantized low-precision student network without KD to obtain a good initialization. Second, Co-studying (CS) phase tries to train a teacher to make it more quantization-friendly and powerful than a fixed teacher. Finally, Tutoring (TU) phase transfers knowledge from the trained teacher to the student. We extensively evaluate our method on ImageNet and CIFAR-10/100 datasets and show an ablation study on networks with both standard and depthwise-separable convolutions. The proposed QKD outperformed existing state-of-the-art methods (e.g., 1.3% improvement on ResNet-18 with W4A4, 2.6% on MobileNetV2 with W4A4). Additionally, QKD could recover the full-precision accuracy at as low as W3A3 quantization on ResNet and W6A6 quantization on MobilenetV2. (See Fig. 1)

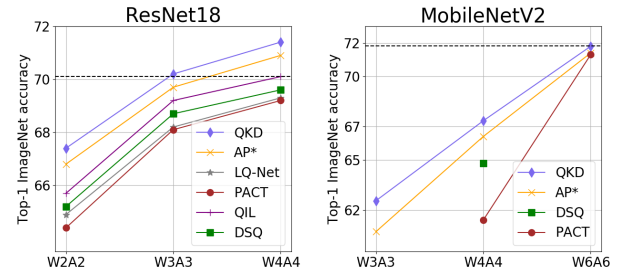


Figure 1: Top-1 accuracy on ImageNet with ResNet-18 and MobileNetV2. Our QKD is compared with various methods such as AP* [30], LQ-Net [50], PACT [6], QIL [18] and DSQ [10]. Dotted line indicates the full-Precision accuracy. More details are in Table 3 and 4.

1. Introduction

Deploying complex DNNs on resource-constrained edge devices such as smartphones or IoT devices is still challenging due to their tight memory and computation requirements. To address this, several research works have been proposed which can be roughly categorized into three folds: weight pruning [12, 11, 27, 9], quantization, [23, 50, 26, 47, 5] and knowledge distillation [15, 20, 48].

Today, a majority of edge devices require fixed-point inference for compute- and power-efficiency. Hence, quantizing the weights and activations of deep networks to a certain bit-width becomes a necessity to make them compatible to edge devices. Recently, hardware accelerators like NPUs (Neural Processing Units), NVIDIA's Tensor Core units and even CIM (Compute-in-memory) devices have targeted support for sub-byte level processing such as 4-bit $\times$ 4-bit or 2-bit $\times$ 2-bit matrix multiply-and-accumulate operations

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